2. For the Web application automate the continuous integration and deployment process using Jenkins.

- Write the code in Visual Studio and push it to your GITHUB repository.
- Modify the code and commit the updates back in GITHUB.
- Using Jenkins configure the job as pipeline and create the stages for the Continuous Integration process.
  - Checkout GITHUB
  - Build using Maven
  - Create Docker image
  - Push to Docker Hub
  - Create Cluster in Kubernetec and Deploy

c. To print rating of a movie

GITHUB REPO LINK: https://github.com/GodStr1k/labfatagile.git
https://github.com/GodStr1k/labfatagile

PIPELINE SCRIPT:

```
pipeline {
   agent any

   environment {
      DOCKER_IMAGE = "godstr1k/web-app:latest"
   }

   stages {

      stage('Clone') {
         steps {
            git branch: 'master',
            url: 'https://github.com/GodStr1k/labfatagile.git'
         }
      }

      stage('Build Docker Image') {
         steps {
            bat "docker build -t %DOCKER_IMAGE% ."
         }
      }

      stage('Push Docker Image') {
         steps {
            withCredentials([usernamePassword(
               credentialsId: 'dockerhub-creds',
               usernameVariable: 'USER',
```

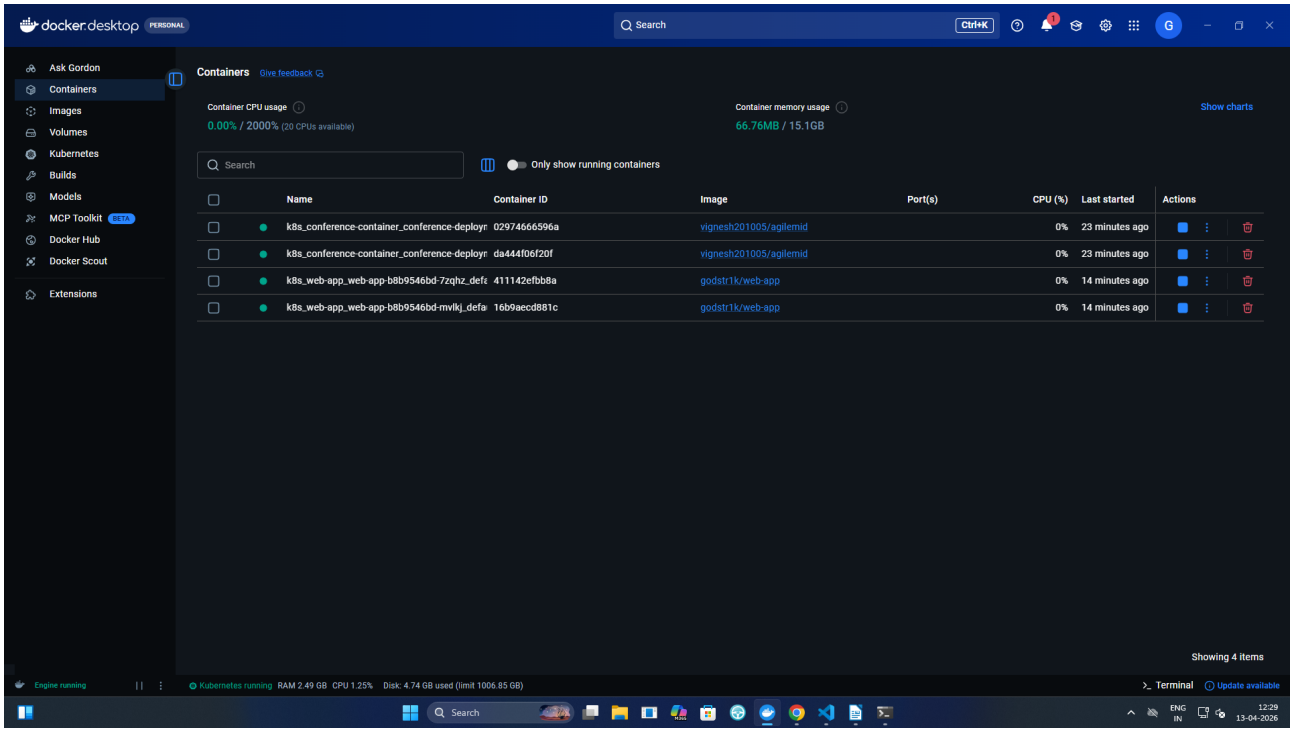
```
                passwordVariable: 'PASS'
            )]) {
                bat """
                docker login -u %USER% -p %PASS%
                docker push %DOCKER_IMAGE%
                """
            }
        }
    }

    stage('Deploy to Kubernetes') {
        steps {
            withCredentials([file(
                credentialsId: 'kubeconfig',
                variable: 'KUBECONFIG'
            )]) {
                bat '''
                set KUBECONFIG=%KUBECONFIG%
                kubectl config set-cluster docker-desktop --insecure-skip-tls-verify=true
                kubectl apply -f deployment.yaml --validate=false
                '''
            }
        }
    }

    stage('Verify') {
        steps {
            withCredentials([file(
                credentialsId: 'kubeconfig',
                variable: 'KUBECONFIG'
            )]) {
                bat '''
                set KUBECONFIG=%KUBECONFIG%
                kubectl get pods
                kubectl get svc
                '''
            }
        }
    }
  }
}
```

STAGE VIEW

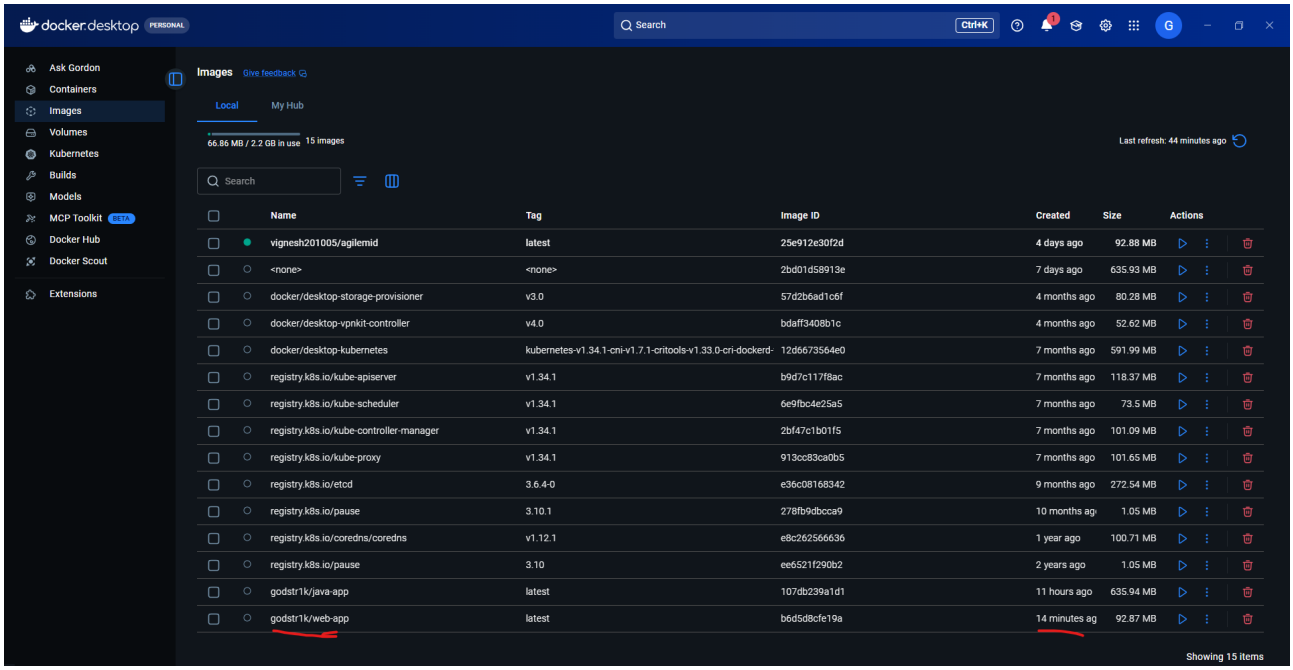
CONSOLE OUTPUT

```
C:\ProgramData\Jenkins\.jenkins\workspace\labfat>set KUBECONFIG=****

C:\ProgramData\Jenkins\.jenkins\workspace\labfat>kubectl get pods
NAME                                    READY   STATUS             RESTARTS        AGE
conference-deployment-64d79ccc88-kgz5h  1/1     Running            1               3d19h
conference-deployment-64d79ccc88-zdk77  1/1     Running            1               3d19h
java-app-5cf585675b-25qvb               0/1     CrashLoopBackOff   9 (3m49s ago)   6d23h
java-app-5cf585675b-lklmk               0/1     CrashLoopBackOff   9 (3m36s ago)   6d23h
web-app-b8b9546bd-7zqhz                 1/1     Running            0               92s
web-app-b8b9546bd-mvlkj                 1/1     Running            0               92s

C:\ProgramData\Jenkins\.jenkins\workspace\labfat>kubectl get svc
NAME                TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)        AGE
conference-service  NodePort    10.104.158.202   <none>        80:30007/TCP   3d19h
java-service        NodePort    10.102.4.132     <none>        80:30008/TCP   6d23h
kubernetes          ClusterIP   10.96.0.1        <none>        443/TCP        16d
web-service         NodePort    10.101.20.183    <none>        80:30010/TCP   1s
[Pipeline] }
[Pipeline] // withCredentials
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

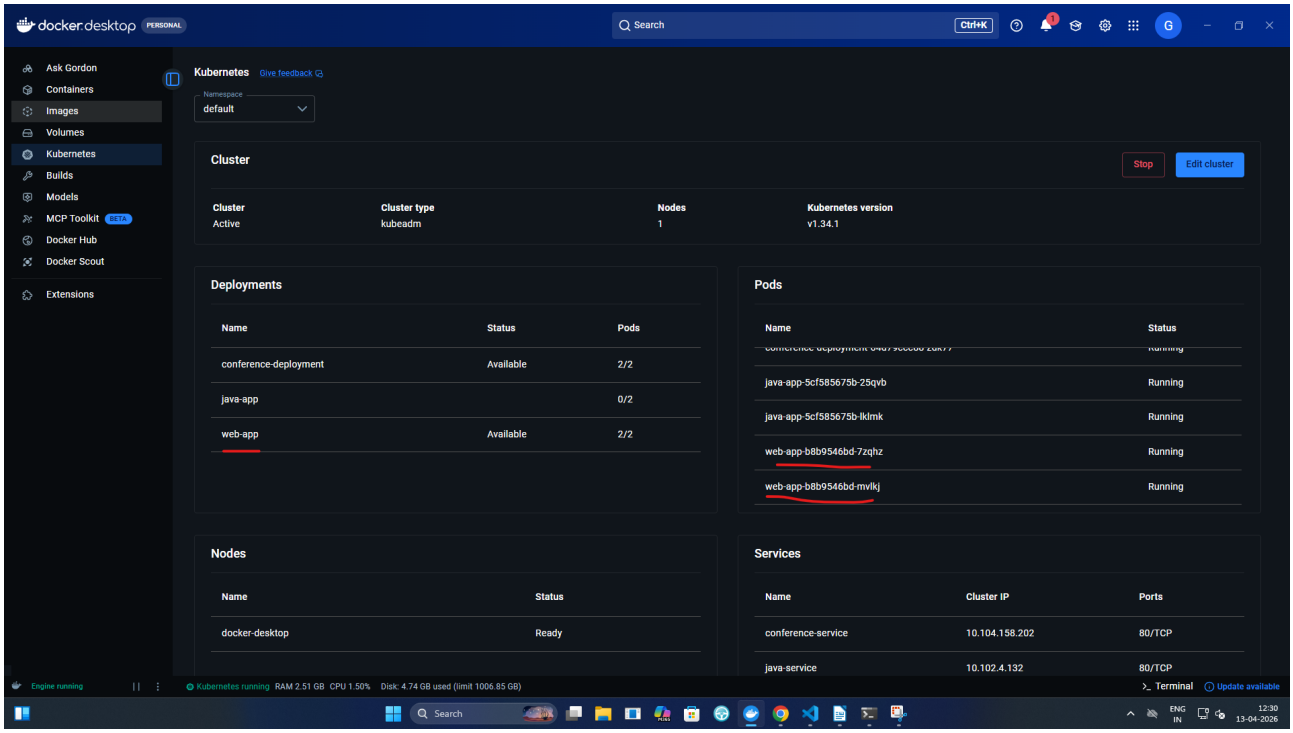
CONTAINER:



IMAGES



KUBERNETES


docker desktop PERSONAL
Search
Ctrl+K
Ask Gordon
Containers
Images
Volumes
Kubernetes
Builds
Models
MCP Toolkit BETA
Docker Hub
Docker Scout
Extensions
Kubernetes
Give feedback
Namespace
default
Cluster
Stop
Edit cluster
Cluster
Active
Cluster type
kubeadm
Nodes
1
Kubernetes version
v1.34.1
Deployments
Name
Status
Pods
conference-deployment
Available
2/2
java-app
0/2
web-app
Available
2/2
Pods
Name
Status
java-app-5cf585675b-25qvb
Running
java-app-5cf585675b-lklmk
Running
web-app-b8b9546bd-7zqhz
Running
web-app-b8b9546bd-mvlkj
Running
Nodes
Name
Status
docker-desktop
Ready
Services
Name
Cluster IP
Ports
conference-service
10.104.158.202
80/TCP
java-service
10.102.4.132
80/TCP
Engine running
Kubernetes running RAM 2.51 GB CPU 1.50% Disk: 4.74 GB used (limit 1006.85 GB)
Terminal
Update available
Search
ENG
IN
12:30
13-04-2026